



In-context learning of pretrained models can be used to directly measure the diversity of LLM generations and human writing.

"I've Seen How This Goes": Characterizing the Diversity of LLM Generations and Human Writing via Progressive Conditional Surprise

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Why measure diversity?

The possibility for diverse outputs is necessary, but not sufficient, for creativity. Researchers routinely need to compare the diversity of outputs across generation processes:

- post-training stages of the same base model (mode collapse)
- decoding strategies (what does temperature trade off against?)
- human–AI hybrid pipelines (would AI use make writing less creative?)

Existing metrics rely on embedding distances or surface-level n -gram statistics, both of which lack a human-like ability to recognize arbitrary patterns.

The idea: in-context learning as a lens

We exploit the in-context learning capability described by Brown et al. (2020), applied here as a measurement lens rather than as a few-shot-prompting setup. As θ sees more responses, its in-context learning picks up on shared modes, stylistic regularities, and topic patterns.

If responses from a policy π are diverse, seeing one should not help θ predict the next, whereas if they are repetitive or constrained to a few modes, conditioning on earlier responses should sharply reduce θ 's surprise at later ones.

Diversity is therefore a property of (responses, prompt, scoring model). The metric measures diversity as θ perceives it.

Takeaways

- A base model's in-context learning detects similarities between arbitrary numbers of responses in a **single forward pass per permutation**.
- No** embedding model, **no** reference corpus, **no** human labels, **no** special-purpose training.
- The same pipeline scores AI samples and human-written sets, and tracks post-training diversity loss on OLMo-2-7B.

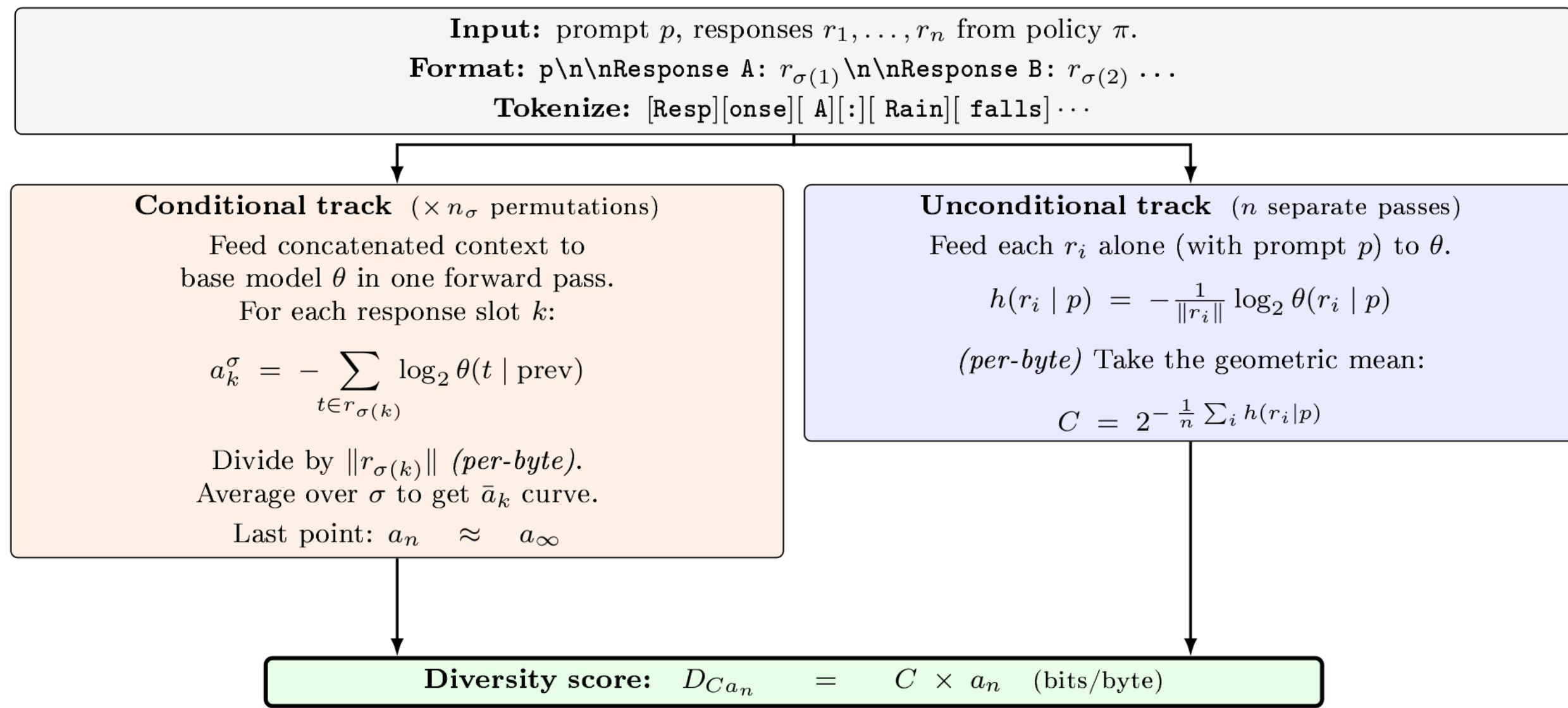


Read the full paper



Code & data

Method: the Decan score $D_{Ca_n} = C \times a_n$



The Decan metric $D_{Ca_n} = C \times a_n$ is a score read off the per-token log-probabilities of a base model θ with one pass per permutation.

Progressive surprise $a_k = -\log_2 \theta(r_k \mid r_{<k}, p)$

Coherence $C = 1/\text{PPL}_\theta(\pi, p)$

Decan $D_{Ca_n} = C \times a_n$ (bits/byte)

Equivalently, $a_n = H(r_n \mid p) - I(r_n; r_{<n} \mid p)$, the response's own surprise minus how much the others make it predictable. High a_n needs r_n to be both surprising and not made predictable by the rest.

The curve alone is fooled by pure noise, since θ can never predict random tokens from one another, so coherence C down-weights incoherent sets and keeps noise from registering as diverse.

We average a_k over uniformly random response orderings, so the curve reflects how θ 's predictions improve with more context, not which response landed in a given slot.

Decan reads as how many bits of surprise per byte remain after θ has learned what it can from the other responses, weighted by output plausibility.

Human benchmark: Tevet & Berant

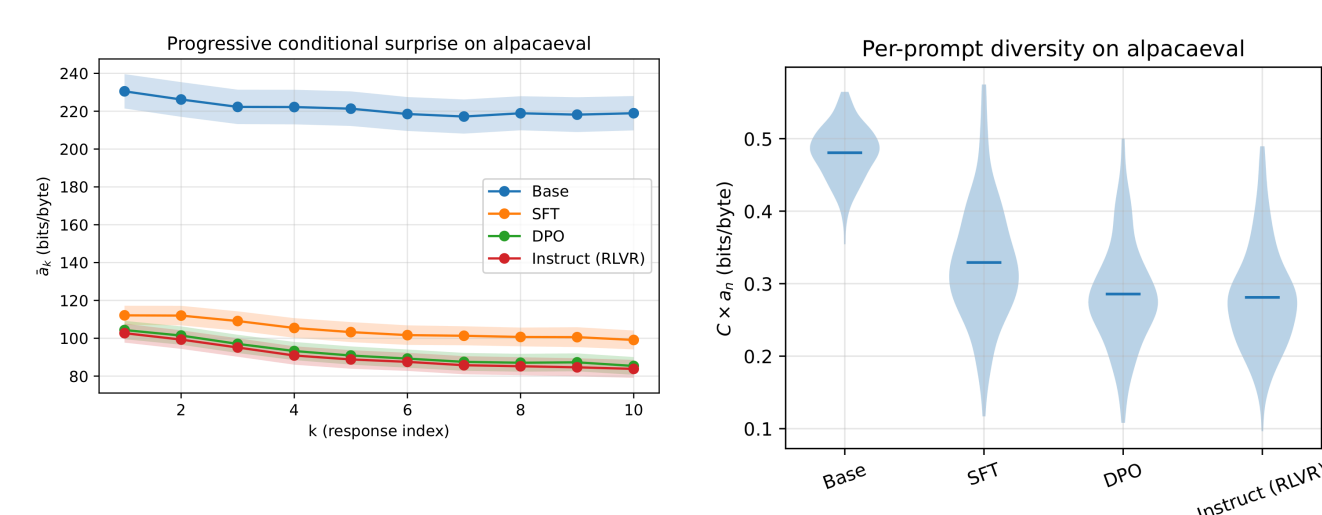
The response sets are written by Mechanical Turk workers, with the high/low diversity label fixed by each dataset's construction protocol.

Across the binary tasks Decan clearly tracks human diversity judgements, between 3.7% and 21.0% behind SentBERT in OCA, reaching close to a sentence-embedding baseline.

Metric	prompt_gen		resp_gen		story_gen	
	ρ	OCA	ρ	OCA	ρ	OCA
<i>ConTest (200, with_hds)</i>						
$C \times a_n$ (ours)	+0.584	0.785	+0.391	0.668	+0.686	0.828
a_n (ours)	+0.444	0.715	+0.274	0.641	+0.387	0.684
C (ours)	+0.214	0.625	−0.001	0.555	+0.247	0.632
SentBERT	+0.682	0.815	+0.591	0.791	+0.770	0.896
BERTsts	+0.646	0.820	+0.463	0.714	+0.601	0.780
distinct- n	+0.333	0.675	+0.346	0.677	+0.573	0.772
<i>McDiv_nuggets (~1K, no_hds)</i>						
$C \times a_n$ (ours)	+0.636	0.785	+0.345	0.649	+0.317	0.634
a_n (ours)	+0.487	0.705	+0.225	0.619	+0.124	0.557
C (ours)	+0.138	0.567	+0.082	0.545	+0.251	0.643
SentBERT	+0.728	0.850	+0.532	0.758	+0.633	0.803
BERTsts	+0.683	0.830	+0.393	0.676	+0.344	0.638
distinct- n	−0.003	0.514	−0.002	0.507	−0.002	0.510
<i>McDiv (full, no_hds, ~2K)</i>						
$C \times a_n$ (ours)	+0.729	0.846	+0.500	0.724	+0.523	0.717
a_n (ours)	+0.617	0.781	+0.432	0.698	+0.402	0.668
C (ours)	+0.138	0.565	−0.007	0.512	+0.171	0.594
SentBERT	+0.796	0.897	+0.678	0.830	+0.753	0.867
BERTsts	+0.780	0.893	+0.614	0.781	+0.571	0.740
distinct- n	+0.476	0.746	+0.517	0.738	+0.535	0.744

Spearman ρ and OCA against human diversity labels. $C \times a_n$ and a_n are ours (Qwen2.5-3B, 50 permutations, per-byte); baselines are from Tevet and Berant. Best per task in bold.

Catches post-training mode collapse



On the OLMo-2-7B post-training pipeline, Decan decreases monotonically across the base, SFT, DPO, and RLVR stages, detecting the type of diversity loss that creative-writing applications care about.

base 0.481 → SFT 0.329 → DPO 0.286 ≈ RLVR 0.281

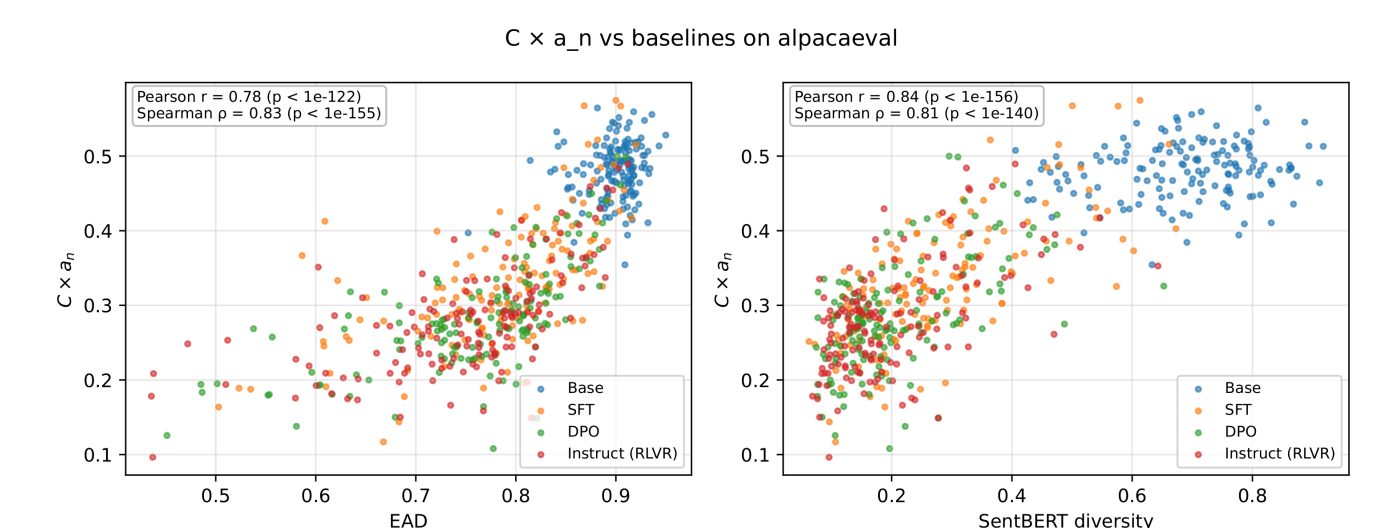
All three pre-registered contrasts are significant after Bonferroni correction, and each later stage's curve lies below the base curve at every $k \geq 2$.

Intended edge-case behavior

Scenario	C	a_n	D_{Ca_n}	GPT-2	Qwen
Pure noise	low	high	low (C)	0.04	0.05
Multi-mode incoherent	low	high	low (C)	0.19	0.29
Multi-mode coherent	high	high	high	0.26	0.19
One-mode	high	low	low (a_n)	0.10	0.09
Mixed	mid	high	mid	0.52	0.41

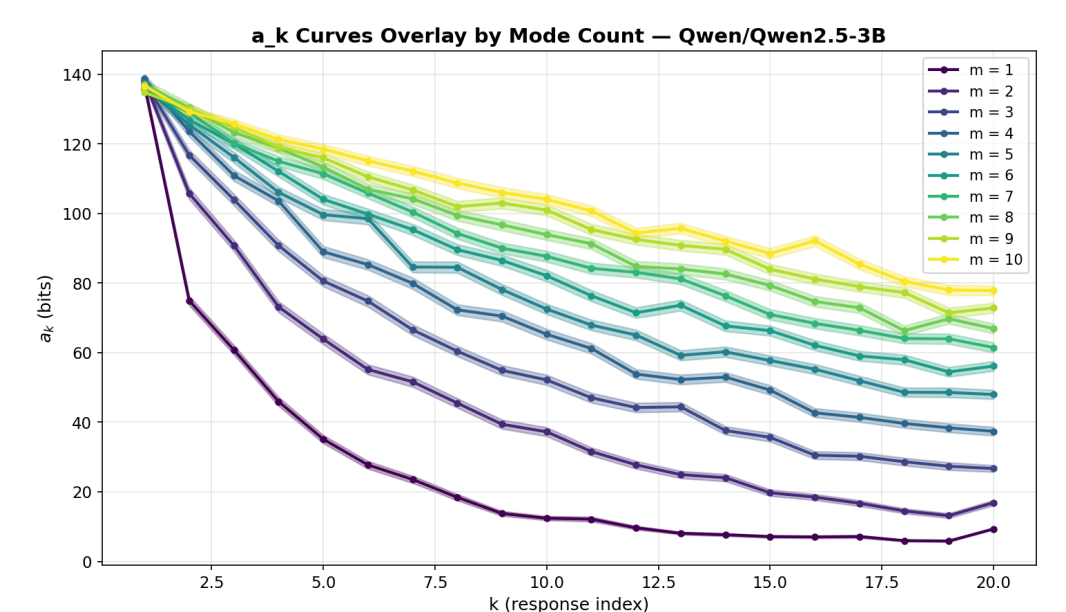
The product suppresses pure noise and incoherent multi-mode via C and one-mode sets via a_n , so multi-mode coherent is the intended (highlighted) winner. Empirically a_n is lower with only 3 modes in 10 responses, so Mixed scores highest and, on Qwen, multi-mode incoherent outranks coherent (see paper).

Agrees with existing metrics



Per-prompt Decan also agrees with the lexical (EAD) and semantic (SentBERT) baselines on the same length-matched subset, confirming the metrics see the same diversity-loss signal.

The curve responds to mode count



With one repeated mode the curve drops sharply toward zero, whereas with many distinct modes it stays high, so a_n rises with the number of distinct modes.

Mode-count scaling on Qwen2.5-3B.